

PAC Functions

A browser supporting PAC provides access to a list of functions as defined in the original Netscape Specification.

Each browser implements PAC in a sandbox, allowing access to only those JavaScript functions required to operate and nothing more. As an example, it isn't possible to access the browser user agent string in a PAC file, a string available to a normal web page.

The functions supported and allowed by the sandbox environment are documented on this page.

The Functions

Click the name of a function in order to view a description and code example(s).

dnsDomainIs

Evaluates hostnames and returns true if hostnames match. Used mainly to match and exception individual hostnames.

Example

```
1 // If the hostname matches google.com or www.google.com
2 // send direct to the Internet.
3
4 if (dnsDomainIs(host, "google.com") || dnsDomainIs(host, "www.google.com"))
5     return "DIRECT";
```

shExpMatch

Will attempt to match hostname or URL to a specified shell expression, and returns true if matched.

Example

```
1 // Any requests with a hostname ending with the extension .local
2 // will be sent direct to the Internet.
3
4 if (shExpMatch(host, "*.local"))
5     return "DIRECT";
```

Example

```
1 // A request for the host vpn.domain.com or any request for a file or folder
2 in the
3 // location http://abcdomain.com/folder/ will be sent direct to the Internet.
4
5 if (shExpMatch(host, "vpn.domain.com") ||
6     shExpMatch(url, "http://abcdomain.com/folder/*"))
7     return "DIRECT";
```

isInNet

This function evaluates the IP address of a hostname, and if within a specified subnet returns true. If a hostname is passed the function will resolve the hostname to an IP address.

Example

```
1 // If IP of requested website falls within IP range, send direct to
2 the Internet.
3
4 if (isInNet(dnsResolve(host), "172.16.0.0", "255.240.0.0"))
5     return "DIRECT";
```

myIpAddress

Returns the IP address of the host machine.

Example

```
1 // If the machine requesting a website falls within IP range,
2 // send traffic via proxy 10.10.5.1 running on port 8080.
3
4 if (isInNet(myIpAddress(), "10.10.1.0", "255.255.255.0"))
5     return "PROXY 10.10.5.1:8080";
```

dnsResolve

Resolves hostnames to an IP address. This function can be used to reduce the number of DNS lookups, e.g. below example.

Example

```
1 // If IP of the requested host falls within any of the ranges specified, send
2 direct.
3
4 if (isInNet(dnsResolve(host), "10.0.0.0", "255.0.0.0") ||
5     isInNet(dnsResolve(host), "172.16.0.0", "255.240.0.0") ||
6     isInNet(dnsResolve(host), "192.168.0.0", "255.255.0.0"))
```

```
7      isInNet(dnsResolve(host), "127.0.0.0", "255.255.255.0"))  
      return "DIRECT";
```

isPlainHostName

This function will return true if the hostname contains no dots, e.g. http://intranet

Useful when applying exceptions for internal websites, e.g. may not require resolution of a hostname to IP address to determine if local.

Example

```
1  // If user requests plain hostnames, e.g. http://intranet/,  
2  // http://webserver-name01/, send direct.  
3  
4  if (isPlainHostName(host))  
5      return "DIRECT";
```

localHostOrDomainIs

Evaluates hostname and only returns true if exact hostname match is found.

Example

```
1  // If the Host requested is "www" or "www.google.com", send direct.  
2  
3  if (localHostOrDomainIs(host, "www.google.com"))  
4      return "DIRECT";
```

isResolvable

Attempts to resolve a hostname to an IP address and returns true if successful. WARNING – This may cause a browser to temporarily hang if a domain isn't resolvable.

Example

```
1  // If the host requested can be resolved by DNS, send via proxy1.example.com.  
2  
3  if (isResolvable(host))  
4      return "PROXY proxy1.example.com:8080";
```

dnsDomainLevels

This function returns the number of DNS domain levels (number of dots) in the hostname. Can be used to exception internal websites which use short DNS names, e.g. http://intranet

Example

```
1 // If hostname contains any dots, send via proxy1.example.com, otherwise send
2 direct.
3
4 if (dnsDomainLevels(host) > 0)
5     return "PROXY proxy1.example.com:8080";
6     else return "DIRECT";
```

weekdayRange

Allows rules to be time based, e.g. only return a proxy during specific days.

Example

```
1 // If during the period of Monday to Friday, proxy1.example.com will be
2 returned, otherwise
3 // users will go direct for any day outside this period.
4
5 if (weekdayRange("MON", "FRI")) return "PROXY proxy1.example.com:8080";
6     else return "DIRECT";
```

dateRange

Allows rules to be time based, e.g. only return a proxy during specific months.

Example

```
1 // If during the period of January to March, proxy1.example.com will be
2 returned, otherwise
3 // users will go direct for any month outside this period.
4
5 if (dateRange("JAN", "MAR")) return "PROXY proxy1.example.com:8080";
6     else return "DIRECT";
```

timeRange

Allows rules to be time based, e.g. only return a proxy during specific hours.

Example

```
1 // If during the period 8am to 6pm, proxy1.example.com will be returned,
2 otherwise
3 // users will go direct for any time outside this period.
4
5 if (timeRange(8, 18)) return "PROXY proxy1.example.com:8080";
6     else return "DIRECT";
```

alert

The `alert()` function is not specified in the original PAC specification, although support was previously supported in several browsers, useful for outputting the value of a variable or result of a function in a manner that is viewable by the end-user and leveraged for troubleshooting PAC file rule issues.

This function is now considered unsupported and non-functional in PAC files.

Example

```
1  // Outputs the resolved IP address of the host in the browser
2  // to end-user or error console.
3
4  resolved_host = dnsResolve(host);
5  alert(resolved_host);
```